Week 5
Inference and Prediction

Applied Statistical Analysis/Quantitative Methods I

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Autumn 2026

Inference: r versus ρ

We observe sample correlation r, but we are often interested in population correlation ρ

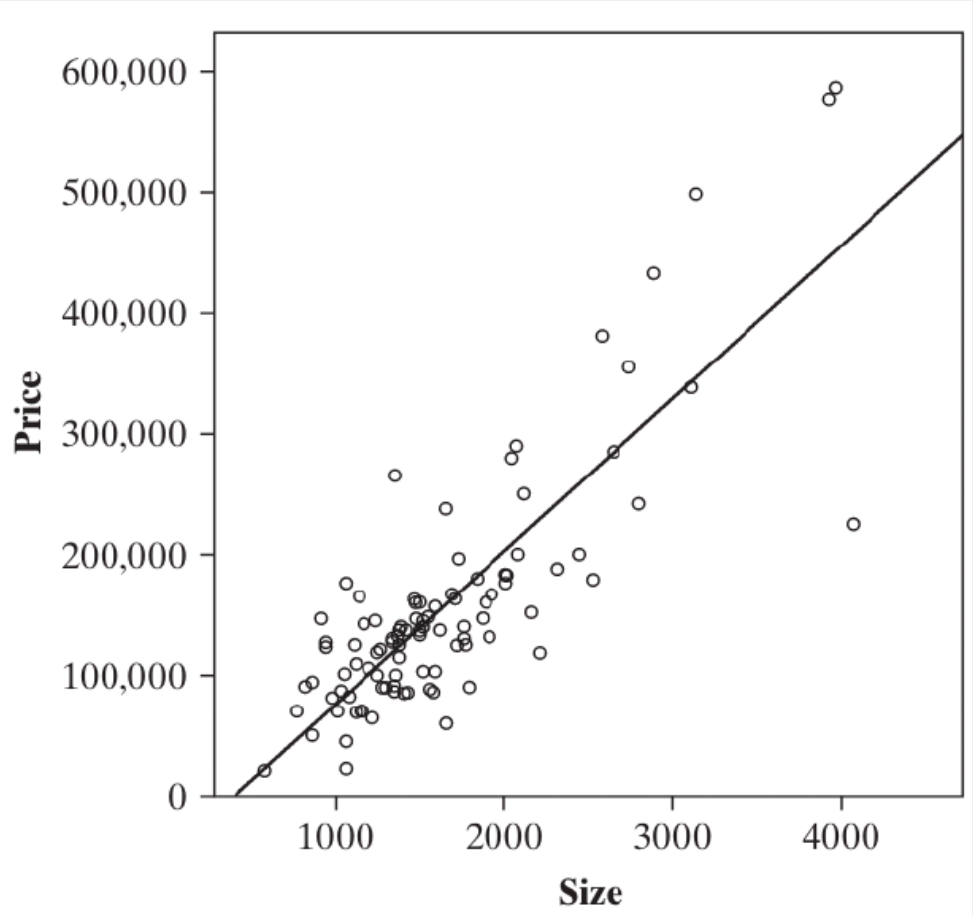
Hypotheses:

$$H_0 : \rho_{XY} = 0 \qquad \text{versus} \qquad H_A : \rho_{XY} \neq 0$$

Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Ex: House Prices and House Size



Suppose we want to assess whether house selling price and house size are correlated

$$r = 0.83378$$

Test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$= \frac{0.834\sqrt{98}}{\sqrt{1-0.834^2}} = 14.95$$

This statistic indicates how likely it would be to observe sample correlation if null hypothesis of no population correlation were true

Regression: How to Pick "Best Line"

Objective of OLS is to minimise residuals

So, let's first try sum of squared errors

$$\text{SSE} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (Y_i - \widehat{Y}_i)^2$$

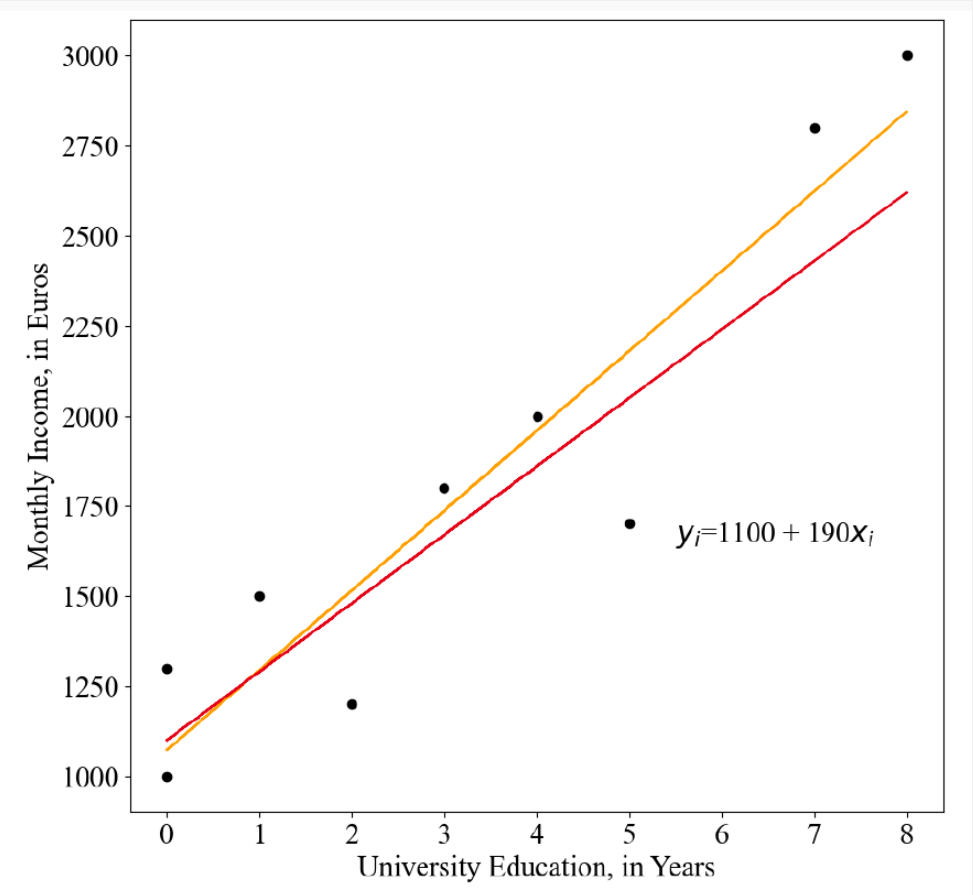
Equivalently,

$$\text{SSE} = \sum_{i=1}^{n} \left(Y_i - \widehat{\alpha} - \widehat{\beta} X_i \right)^2$$

Comparing Candidate Lines

Two possible regression lines may fit the same data differently

Better line is the one with smaller total squared residuals (i.e., above and below regression line):



Ex:

- Residuals of red = 612,800
- Residuals of orange = 490,588

Therefore,

$$\text{SSE}_{\text{red}} > \text{SSE}_{\text{orange}},$$

so orange line provides better fit

Why exactly is this the "best line"?

Gauss–Markov Theorem

- Unbiased: An estimator is unbiased if its expected value is identical to the population parameter:

$$E[\widehat{\theta}] = \theta$$

- Best: In sense that it has lowest variance among all linear unbiased estimators
- Linear: Estimator is a linear function of sample data points
- Under standard linear model assumptions

OLS is BLU (Best Linear Unbiased Estimator)

What were those assumptions again?

- ϵ_i is normally distributed $\rightarrow$ needed for inference
- $E(\epsilon_i) = 0$, no bias $\rightarrow$ violated if error is not random
  - Possibly correlated with omitted variable
- ϵ_i has constant variance σ^2 (Homoscedasticity $\rightleftarrows$ Heteroscedasticity)
- $Cov(\varepsilon_i, \varepsilon_j) = 0, \forall i \neq j$
- Parametric linearity

Good source: Kellstedt and Whitten (2018), 190–194

Getting the Coefficients

The least-squares point estimates are:

$$\widehat{\alpha} = \bar{y} - \widehat{\beta}\bar{X}$$

and

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Estimates provide line that minimises sum of squared errors

Regression Diagnostics

Regression diagnostics help us assess whether model adequately represents data

- Patterns in residuals may indicate systematic influences on Y that model has not captured
- Patterns may indicate that functional form of model is inappropriate
- Residual patterns may also indicate that errors do not have constant variance

By construction, OLS residuals sum to zero:

$$\sum_{i=1}^{n} \left(y_i - \widehat{\alpha} - \widehat{\beta}_1 x_i \right) = \sum_{i=1}^{n} (y_i - \widehat{y}_i) = \sum_{i=1}^{n} e_i = 0.$$

This does not mean that model is correct!!!

Residual Plots

Residual plot is a scatterplot of regression residuals against:

- explanatory variable X, or
- predicted values $\widehat{Y}$

Ideally, residuals should look as if they were generated by pure chance, with no systematic pattern

Inference About β: The Logic

OLS provides an estimate $\widehat{\beta}$, but this estimate is based on a sample

The inferential question is:

- Does evidence suggest that true slope β differs from zero?

Our usual hypotheses are

$$H_0 : \beta = 0 \qquad \text{versus} \qquad H_A : \beta \neq 0$$

The Test Statistic

Under null hypothesis $H_0 : \beta = 0$, our test statistic is:

$$t = \frac{\widehat{\beta} - 0}{SE(\widehat{\beta})}$$

We then compare $|t|$ with critical value

$$t_{n-2,\,1-\alpha/2}$$

or calculate the corresponding p-value

Interpreting Our Test Statistic

- If $|t|$ is large, the estimate is far from zero relative to its standard error
- A large absolute t-statistic produces a small p-value
- A small p-value provides evidence against H_0

We may reject H_0 and conclude that X has a statistically significant linear association with Y

Standard Errors and Sampling Uncertainty

Standard error of $\widehat{\beta}$ measures how much estimated slope would vary across repeated random samples:

$$SE(\widehat{\beta}) = \frac{\widehat{\sigma}}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

Residual standard deviation is

$$\widehat{\sigma} = \sqrt{\frac{\text{SSE}}{n-2}} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{n-2}}$$

What Improves Precision?

- A smaller $SE(\widehat{\beta})$ means that β is estimated more precisely
- Precision improves when:
  1. residuals are smaller, indicating a better model fit
  2. sample size n is larger
  3. X has greater variation

Confidence Interval for β

A confidence interval for the slope is:

$$\widehat{\beta} \pm t_{n-2,\,1-\alpha/2} SE(\widehat{\beta})$$

A 95% confidence interval means that, if sampling process were repeated many times, approximately 95% of resulting intervals would contain true value of β

- But we don't know which times!!!

Ex: Confidence Interval

Suppose

$$\widehat{\beta} = 221.6, \qquad SE(\widehat{\beta}) = 35.0$$

95% confidence interval is

$$[152.2, 291.0]$$

Substantively, each additional year of education is associated with an increase of approximately 152 to 291 euros in expected income

Interpreting Significance and Magnitude

Statistical significance concerns whether a relationship is distinguishable from zero given sampling uncertainty

Substantive magnitude concerns whether the size of $\widehat{\beta}$ is meaningful in context

- In large samples, even small effects can be statistically significant
- In small samples, large but noisy effects may fail to reach statistical significance

What to Report

Regression results should normally include:

- Point estimate $\widehat{\beta}$
- Standard error
- p-value or confidence interval
- Substantive interpretation connected to research question

Putting It All Together

1. Estimate coefficients $\widehat{\alpha}$ and $\widehat{\beta}$ using OLS
2. Check model assumptions and inspect residual plots
3. Compute standard errors
4. Perform a hypothesis test and report p-value:

$$t = \frac{\widehat{\beta}}{SE(\widehat{\beta})}$$

5. Alternatively: Compute and interpret confidence intervals
6. Translate statistical results into substantive conclusions

Class business

- Read required (and suggested) online materials
- Finish Problem set #2!